EFU FOSSIL FUELS LICENCE v1.0

Experimental EFU-based Oil & Natural Gas Extraction/Processing Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for fossil fuel extraction, refining, and distribution.

NOT a legal extraction concession, environmental permit, or EU ETS substitute, but an open research protocol measuring fossil energy physical footprint in human-scale EFU metrics (energy, water, materials, entropy, carbon).

I. Core Concepts

1. **EFU-Fossil (EFU-FOSSIL)**
   EFU framework application for fossil fuels:
   - **EFU-H_FOSSIL**: primary energy input (extraction/refining energy)
   - **EFU-W_FOSSIL**: water (fracking, cooling, desalination)
   - **EFU-M_FOSSIL**: materials (cement, steel, chemical catalysts)
   - **EFU-S_FOSSIL**: entropy (heat loss, combustion products)
   - **EFU-C_FOSSIL**: end-use CO_2 (combustion + methane leakage)
2. **Fossil Sovereignty Gap (SS_FOSSIL)**
   SS_FOSSIL = TEFU_FOSSIL_extraction – Regional Energy Carrying Capacity (RECC_FOSSIL)
   Measures if fossil extraction/refining stays within regional sustainable energy regeneration frame.
3. **MROI_FOSSIL**
   MROI_FOSSIL = THI_FOSSIL / TEFU_FOSSIL
   THI_FOSSIL = energy supply security + industrial productivity + export revenues.

II. EFU Fossil Licence – Research Status

Documents EFU-audited extraction/refinery operations.
Does NOT replace extraction rights, safety permits, or carbon tax systems.
Enables oil fields, refineries, gas processing benchmarking.

III. Metrics and Thresholds

1. EFU Components (per extracted barrel or GJ)

EFU-H_FOSSIL (Energy)
EFU-H_FOSSIL = extraction_energy_MWh / 12.88 / extracted_energy_GJ
Example: 1 barrel oil requires 0.1 MWh → 0.1/12.88/42GJ = 0.000185 EFU/GJ

EFU-W_FOSSIL (Water)
EFU-W_FOSSIL = fracking_water_m3 / 18.25 / extracted_energy_GJ
Example: US shale = 15 m^3/barrel → 15/18.25/42GJ = 0.0195 EFU/GJ

EFU-M_FOSSIL (Materials)
EFU-M_FOSSIL = cement_steel_kg / 30 / extracted_energy_GJ

TEFU_FOSSIL = 0.8×EFU-H + 1.5×EFU-W + 1.0×EFU-M + 1.2×EFU-S + 2.0×EFU-C
(high EFU-C weight due to end-use CO_2)

2. Regional Energy Carrying Capacity (RECC_FOSSIL)

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RECC_FOSSIL = Sustainable_primary_energy_TWh / Population / 12.88 MWh/person/year
Example: Saudi Arabia = 1,200 TWh / 36M / 12.88 = 2.58 EFU_FOSSIL/person/year
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IV. Licence Categories (EFU-FOSSIL Labels)

Level	SS_FOSSIL	MROI_FOSSIL	Label	Interpretation
1	≤ 0	≥1e-3	**EFU-FOSSIL SOVEREIGN**	Sustainable extraction
2	0 < ≤50k EFU	≥1e-4	**EFU-FOSSIL BALANCED**	Acceptable
3	50k < ≤200k EFU	≥1e-5	**EFU-FOSSIL DEFICIT**	Overloaded
4	>200k EFU	<1e-5	**EFU-FOSSIL CRITICAL**	Metabolic predator

V. Regional RECC_FOSSIL Calibrations

Country	Primary Energy (TWh/year)	Population (M)	RECC_FOSSIL (EFU/person/year)
SAUDI ARABIA	1,200	36	2.58
USA	25,000	340	0.57
RUSSIA	7,500	144	0.41
CHINA	35,000	1,410	0.19
EU	12,000	448	0.21

VI. Example: Saudi Arabian Oil Field

Input (annual field):

- Production: 100M barrels (4,200 PJ)
- Extraction energy: 10,000 MWh
- Fracking water: 1.5B m³
- Staff: 5,000 workers
- Region: Saudi Arabia (RECC_FOSSIL = 2.58 EFU/person/year)

Calculation:

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EFU-H_FOSSIL = 10,000/12.88/4,200,000 = 0.000185 EFU/GJ
EFU-W_FOSSIL = 1,500,000/18.25/4,200,000 = 0.0195 EFU/GJ
EFU-C_FOSSIL = 400M tCO2/4/4,200,000 = 0.0238 EFU/GJ (combustion)

TEFU_FOSSIL_total ≈ 180,000 EFU/year
Allocated = 5,000 × 2.58 = 12,900 EFU/year
SS_FOSSIL = 180,000 – 12,900 = +167,100 EFU/year (Level 3: DEFICIT)

Result: EFU-FOSSIL DEFICIT
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VII. Implementation Notes

1. **EU ETS complement**: TEFU_FOSSIL basis for carbon pricing
2. **Data sources**: Tier 1 = extraction logs, water meters; Tier 2 = IEA statistics
3. **Python ready**: EFU-Fossil module follows EFU-Fish/Forest logic